UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2022

Roll No. ....

Paper: Analytical Chemistry | Course Code: CHEM-319 | Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

i. Define ion exchange capacity and give its formula? IEC
ii. Why is it necessary to keep resin hydrated during Ion exchange chromatography? IEC
iii. What is the principle of gel chromatography? GC
iv. Differentiate between simple and multiple extraction system? SE
v. How does ion exchange chromatography help in softening of hard water? IEC
vi. What is separation efficiency of metal chelates? SE
vii. Why is solid phase extraction better than liquid phase extraction? SE
viii. What is Basic principle of AAS? AAS
ix. What are the characteristics of air acetylene flame? AAS
x. Why low temperature is used for alkali and alkaline earth metals? AAS
xi. What is the role of complexing agents in metal extraction? SE
xii. How is solvent extraction governed by partition law? SE
xiii. How does electrophoresis help in gene and genome analysis? EP
xiv. What are the characteristics of anion exchange resins? IEC
xv. What causes the red feather observed in the Nitrous oxide acetylene flame? AAS

Q.2 Write brief answer of following. (6x5=30)

a) Write down the working and characteristics of hollow cathode lamp. AAS
b) Briefly explain the process of packing of column and separation in gel chromatography? GC
c) Explain countercurrent process in solvent extraction. SE
d) Explain capillary zone electrophoresis. EP
e) Describe process of flow injection analysis in solvent extraction. SE
f) What is basic principle of Gel chromatography? Briefly explain different types of gels used. GC

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B.S. 4 Years Program / Sixth Semester – 2020

Roll No. ....

Paper: Analytical Chemistry
Course Code: CHEM-319 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2- Attempt all Short questions (10x2=20)

i. How are samples introduced in capillary electrophoresis system? EP
ii. Define and explain the Efficiency of solvent extraction? SE
iii. Define electrophoretic mobility? Derive its unit. EP
iv. What is the role of complexing agent in metal extraction? SE
v. Define Nernst distribution law? Give an example. SE
vi. Why atomic absorption spectroscopy is limited to metals only? AAS
vii. Why gel chromatography is also known as size exclusion chromatography? GC
viii. What are ion exchangers? Give one example of each type? IEC
ix. What are the limitations of flame photometry? FES
x. What is the difference between atomic absorption and flame emission spectroscopy? AAS

Q 3. Long Questions (5x6 =30)

a) Explain briefly all steps involved in flow injection analysis? SE
b) What are different types of ion exchange resins? Explain cross linkage in ion exchange resin with the help of an example. IEC
c) Explain Zeta potential in detail on the basis of electroosmotic flow? EP
d) Give the principle and applications of Gel chromatography? GC
e) Discuss different events occurring in flame emission spectroscopy with the help of a diagram? FES
f) Discuss Hydride generation method in detail specially emphasizing on its principle? AAS

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2021

Roll ....

Paper: Analytical Chemistry
Course Code: CHEM-319 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. An[illegible]wing short questions. (10x2=20)

1. Compare[illegible]ne and Nitrous oxide-acetylene flames in AAS? AAS
2. Explain chemical interference with the help of an example? AAS
3. What is the principle of solid phase extraction? SE
4. What do you meant by the term percent extraction? Explain SE
5. What are the characteristics of anion exchange resin? IEC
6. Define electroosmotic flow? Write down its mathematical formula? EP
7. Why it is not suitable to pack the column with dry gel in Gel chromatography? GC
8. How does ion exchange chromatography helps in softening of Hard water? IEC
9. What is the difference between distribution coefficient and distribution ratio in solvent extraction? SE
10. Why low temperature flame is used for the alkali and alkaline earth metals in FES? FES

Q.3. Answer the following questions. (3x10=30)

1(a): Explain the construction and working of Graphite furnace with the help of diagram? AAS(5)

(b) Write down the applications of flame emission spectroscopy? (5) FES

2(a) Explain different types of factors affecting ion exchange chromatography? (5) IEC

Size
Valence
conc
Nature

(b) Explain capillary zone electrophoresis in detail ? (5) EP

3(a) Explain countercurrent process in solvent extraction? (5) SE

(b) Gi[illegible] account on gel chromatography? (5) GC

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Mobile
Detector
Column

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Analytical Chemistry **TIME ALLOWED: 2 Hrs. & 45 Mints.**

Course Code: CHEM-319 Part – II **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Section I

Q.2- Attempt all Short questions (2x10=20)

(i) How do Doppler Effect and pressure broadening affect the transitions in atomic spectroscopy? AAS

(ii) What are the characteristics and advantages of weak base anion resins? IEC

(iii) How are samples introduced in capillary electrophoresis system? EP

(iv) Describe the term percent extraction and separation factor in solvent extraction. SE

(v) How electro-osmotic flow can be altered in capillary electrophoresis? EP

(vi) Describe the role of masking and oxidation state in increasing the selectivity of metal extractions in solvent extraction. SE

(vii) Compare air-acetylene and nitrous oxide-acetylene flames in AAS. AAS

(viii) What do you know about chemical interferences in atomic spectroscopy? AAS

(ix) Write down the principle of solid phase extraction SE

(x) Describe nebulization in flame emission spectroscopy. FES

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Section II

Attempt all questions (3 x 10 = 30)

Q.3 (a) Explain the construction and working of graphite furnace. AAS (5)

(b) Discuss different components of flow injection analysis. SE (5)

Q.4 (a) Discuss the factors affecting ion exchange separations. IEC (5)

(b) Give an account of capillary zone electrophoresis. EP (5)

Q.5 (a) Discuss strong acid cation exchange resins and weak acid cation exchange resin. (5) IEC

(b) What is basis of flame emission spectroscopy? Explain various events taking place during this process FES (5)

Roll No. ....

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2019

Paper: Analytical Chemistry | Time: 2 Hrs. 45 Min. Marks: 50

Course Code: CHEM-319 Part – II

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

SUBJECTIVE

Section I

Q.2- Attempt all Short questions (2x10=20)

(i) What is isoelectric point? IEC
(ii) Differentiate between Flame photometry and atomic absorption spectroscopy. AAS
(iii) Write examples of strong acid cationic resins. IEC
(iv) What do you mean by percent extraction? SE
(v) Write principle of atomic absorption of spectroscopy. AAS
(vi) Describe significance of Gel chromatography.
(vii) What do you know about electro osmotic flow? EP
(viii) How does ion exchange chromatography help in softening of hard water? IEC
(ix) What is solid phase extraction? SE
(x) What is the role of complexing agents in metal extraction? SE

Section II

Attempt all questions. (5x6=30)

Q 3:

a) Explain all steps involved in flow injection analysis. SE
b) Discuss different types of resins IEC
c) How are metals extracted by solvent extraction? Explain. SE
d) Write applications of flame emission spectroscopy. FES
e) Describe hydride generation in flameless method in AAS. AAS
f) Discuss factors affecting ion exchange chromatography. IEC

2015

1) cross linkage in ion exchange chromatography? IEC

2,3,4 repeat in 2014

5) order of species saperated in normal capillary zone electrophoresis? EP

6) effect of electrophoretic mobility on the saperation in CZE? EP

7,8 repeat in 2014

9) Gel chromatography as size exclusion? GC

10) Burner System in Flame photometry? FES

2017

1) what is Synergistic Extraction? SE
2) What is Electrosmotic flow ? EP
3) Characteristics of Cation Exchange resins ? IEC
4) Multiple extraction in solvent extraction? SE
5) purpose of internal standards?
6) write the difference between Distribution ratio and Distribution co-efficiant ? SE
7) Spectral interference? AAS
8) What is Gel filteration ? GC
9,10 repeat in previous..

(2014)

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short questions

1) why low temperature flame used for alkali and alkaline earth metals? FES

2) write the function of flame in photometry? FES

3) what is electrophoretic moblity and electrosmosis mobility? EP

4) why AAS limited to Metals ? AAS

5) write limitations of flame photometry? FES

6) why ion exchange usually swells in water? IEC

7) parking of column with dry gel? GC

8) distribution ratio at constant temperature? SE

9) write distribution law? SE

10) what is ion exchange chromatography? IEC

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